THE UNITED REPUBLIC OF TANZANIA

PRESIDENT'S OFFICE

REGIONAL ADMINSTRATION AND LOCAL GOVERNMENT

KASULU TOWN COUNCIL

FORM TWO PRE-MOCK EXAMINATION

CODE: 031 **PHYSICS**

Time: 2:30 Hours **Friday, 23.05.2025**

INSTRUCTIONS

1. This paper consists of section A, B and C with the total of ten (10) questions.
2. Answer all questions in the space provided.
3. All writing must be in blue or black ink except drawings which must be in pencil.
4. Communication devices and any unauthorized materials are not allowed in Examination room.
5. Write your Examination number at the top right conner of every page.
6. Where necessary the following constant may be used.
   (i) Acceleration due to gravity, $g=10m/s^2$
   (ii) Pie =3.14

MsomiBora.com

SECTION A (15%)

1. For each of the items (i) to(x) choose the correct answer among the given alternative and write its latter beside the item number in the answer.
   (i) Results obtained from physics experiment can form.
   (A) Scientific laws (B) Scientific theories
   (C) Scientific principles (D) Scientific procedures
   (ii) A force which prevents a body to slide is called.
   (A) Stretching force (B) Restoring force
   (C) Frictional force (D) Compressional force
   (iii) In a solid state the force of attraction between molecules is greater because particles are..
   (A) Moved so randomly (B) Somehow apart
   (C) NOT closely packed together (D) Closely packed together
   (iv) Which of the following materials can be magnetized strongly?
   (A) Nickel and copper (B) steel and Brass
   (C) Cobalt and iron (D) cobalt and copper
   (v) A current of 0.2A flows through a resistor of 4 ohms The potential difference across a resistor is
   (A) 20 V (B) 0.8V (C) 0.05V (D) 8V
   (vi) How can real image be distinguished from a virtual image?
   (A) Real image is inverted while virtual image is upright.
   (B) Real image is upright while virtual image is inverted.
   (C) Virtual image is formed by convergent rays while real image is formed by divergent rays.
   (D) Real image is formed by convergent rays while virtual image is formed by divergent rays.
   (vii) A bus carrying a heavy load on its top most carries can easily overturn when negotiating a corner because its
   (A) Equilibrium is neutral
   (B) Center of gravity is low
   (C) Centre of gravity is high
   (D) Centre of gravity is normal

(viii) A person standing in a bus which starts to move forward suddenly tends to fall backwards.
This tendency obeys; -
(A) Newton's first law of motion
(B) Newton's second law of motion
(C) Newton's third law of motion
(D) The principle of moments
(ix) The boiling point of pure water at the sea lever is.
(A) 0ºF (B) 100ºF (C) 200ºF (D)212ºF
(x) Motor vehicle tyres are made up of grooved rubber in order to.
(A) Decrease speed of the vehicles on the road for safety purposes
(B) Increase friction between tyres and road for safety braking
(C) Increases speed of the vehicles on the road
(D) Decrease stability of the tyres on the road

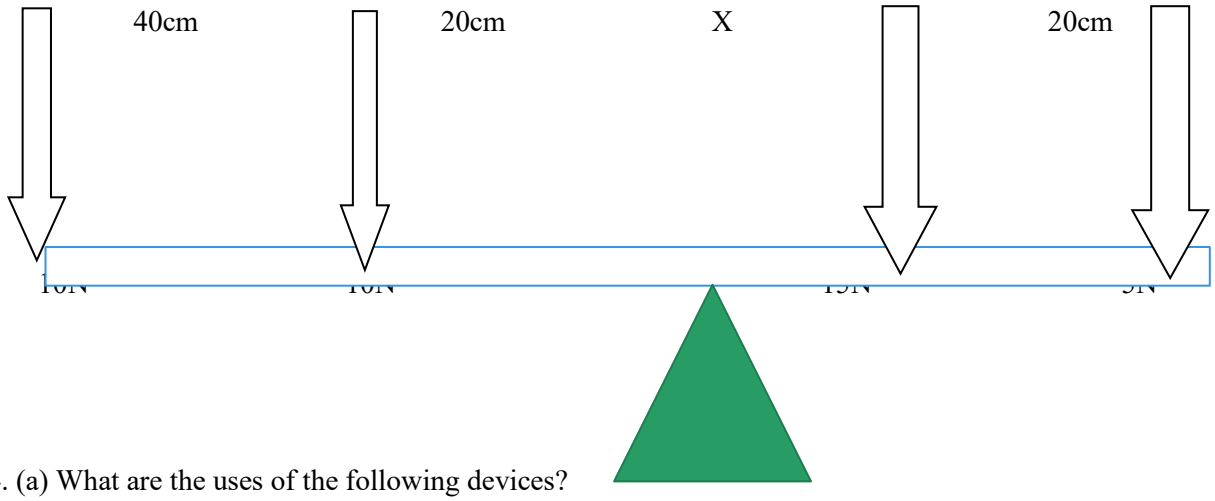
2. Match the item in LIST A with a correct response in LIST B by writing a letter of the correct response below the number of the corresponding item in list A in the table provided.

LIST A	LIST B

(i) Ratio of number of teeth in a driven wheel to the number of teeth in driving wheel. (ii) The force used to operate a machine. (iii) An examples of third-class lever (iv) It is used to lift heavy weights using a small effort. (v) A force that causes the efficiency of a machine to be less than 100%.	(A) Knife (B) Lever (C) Inclined planes (D) Friction (E) Wheel barrow (F) Effort (G) Velocity ratio.

SECTION B(60%)

3. (a) State two conditions for a body to be equilibrium.
   (b) Why is advised to open the cap of the bottle with a bottle opener than with your fingers.
   (c) Study the diagram below and determine the value of x and hence the length of the bar as shown in figure below.



4. (a) What are the uses of the following devices?

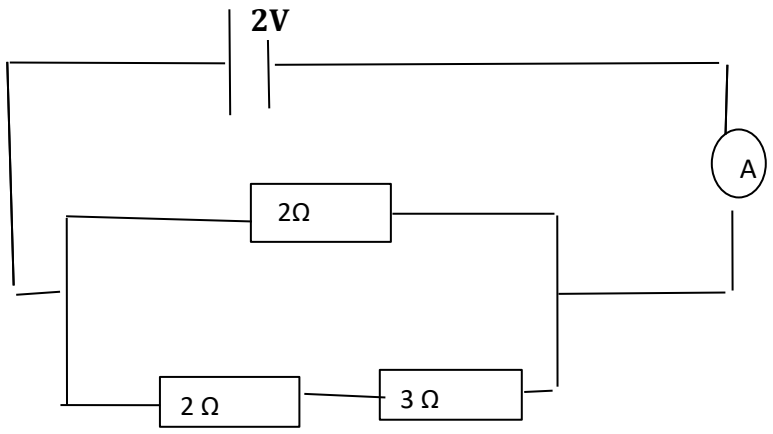
   (i). Manometer. ----------
   (ii). U- Tube. ----------
   (iii). Barometer. ----------
   (iv). Here's apparatus. ----------

   (b) Explain why does a big elephant manage to walk confutable in muddy soil without sinking while human sink easily?
   (c) Draw a well labelled diagram which demonstrates that liquid pressure depends on depth.
5. (a) Explain how area of plates and distance between plates affect the capacitance of the capacitor.
   (b) A capacitor of capacitance 250μF is allowed to charge until the potential difference between its plates is 10V. How much charge accumulated on the plates during the charging process?
6. (a) A form two students performed an experiment on determination of the acceleration due to gravity by using an iron metal ball. Briefly describe the energy changes when a bomb of a simple pendulum swings from one side to another.
   (b) A pump raises 1000kg of water through the height of 30m in 100sec. What is power developed by the pump.

7. (a) State Archimedes' principles.
   (b) Explain relative density of a solid and give SI unit?
   (c) The mass of a density bottle is 15g. When is full filled with a fluid of density 1.2g/cm³ its mass is 51g . Find the volume of liquid.
8. (a) State Newton's second law of motion.
   (b) Explain two examples of application of Newton's third law of motion.
   (c) A body of mass 8kg moving with velocity of 20m/s collides with another body of mass 4kg moving with a velocity of 10m/s in the same direction. The velocity of 8kg body is reduced to 15m/s after collision. Calculate the final velocity of the 4kg body.

SECTION C (25%)

9. (a) Explain how an inclined plane makes it easier to move a load from a lower to a higher position.
   (b) Draw a diagram of combined pulley system with velocity ratio of 4.
   (c) A pulley system is made up of 8 pulleys. An effort of 200N is applied the pulley system. If the pulley has an efficiency of 80%. Find
      (i) Mechanical advantage of the pulley.
      (ii) Maximum load that can be raised by the effort.
10. (a) Explain how an ammeter and voltmeter are connected in a circuit.
    (b) In the circuit shown in figure below the battery and an ammeter have negligible internal resistance. Determine the ammeter reading



THE UNITED REPUBLIC OF TANZANIA
PRESIDENT OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
ILALA TOWN COUNCIL
FORM TWO PRE-MOCK JOINT EXAMINATION
PHYSICS
MARKING SCHEME MAY 2025

1

(i)	(ii)	(iii)	(iv)	V	VI	VII	VIII	IX	X
B	C	D	B	B	A	C	A	D	B

1 marks @

2

LIST A	(I)	(II)	(III)	(IV)	V
LIST B	G	F	A	B	D

1 marks @

03 (a)(i) The net force on the object must be zero.
(ii) The System (body) Must have an acceleration of $0 m/s^2$

OR

(i) The sum of the forces in one direction must be equal to the sum of the forces in the opposite direction
(ii) The sum of clockwise moment should be equal to the sum of anticlockwise moment. (3 marks)

(b) Because due to the addition perpendicular distance arise from the opener. (2 marks)

SOLUTION

MsomiBora.com

C

From principle of moment

The Sum of clockwise moment = The sum of anticlockwise moment. — 1 marks

The value of x and the length of bar are requires

NOW

CM = AM

$10N(40cm+20cm) + 10N \times 20cm = 15N \times x + (20cm+x)5N$ — 2 marks

$10N \times 60cm + 10N \times 20cm = 15Nx + 100cmN + 5Nx$

$15Nx + 100cmN + 5Nx$

3(c) $800Ncm = 20Nx + 100cmN$

$800Ncm - 100Ncm = 20Nx$

$\frac{700Ncm}{20N} = \frac{20Nx}{20N}$

$x = 35cm$ — 2 marks

Value of $x = 35cm$

AGAIN.

The length of bar is required

Now Length of bar $= 40cm + 20cm + 35cm + 20cm$

$= 115cm$ — 2 marks

Length of bar is 115cm.

∴ The value of X is 35cm and length of bar is 115cm

4(a)(i) Manometer

is a device for measuring gas pressure (1 mark)

(ii) U-tube

is a device used for measuring pressure of a gas (1 mark)

(iii) Barometer

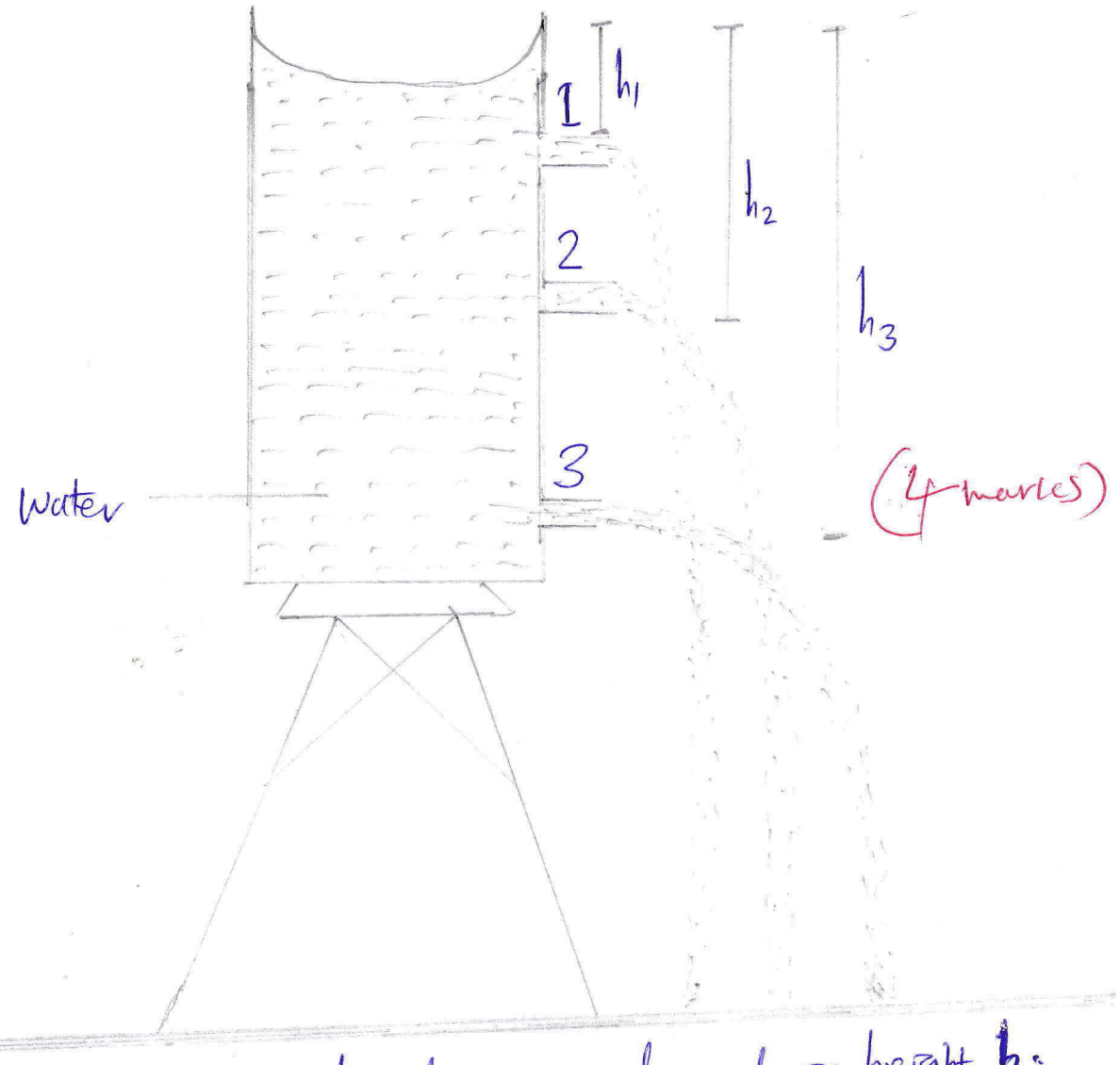
is an instrument for measuring atmospheric pressure (1 mark)

(iv) Hare's apparatus

is an instrument for comparing densities of two liquids. (1 marks)

(b) Because, the wide area of an elephant's feet ensures that the pressure due to its weight is low. On the other hand, the small area of a human foot may cause more pressure on the mud and lead to sinking. (03 marks)

4 (c)



(4 marks)

Demonstrating that liquid pressure depends on height h.

5 (a) Area and distance between the plates

The capacitance increases with closeness of the plates ($C \propto \frac{A}{d}$). (3 marks)

(b) SOLUTION

Data

Capacitance (C) = 250 μF = 250×10^{-6} F.

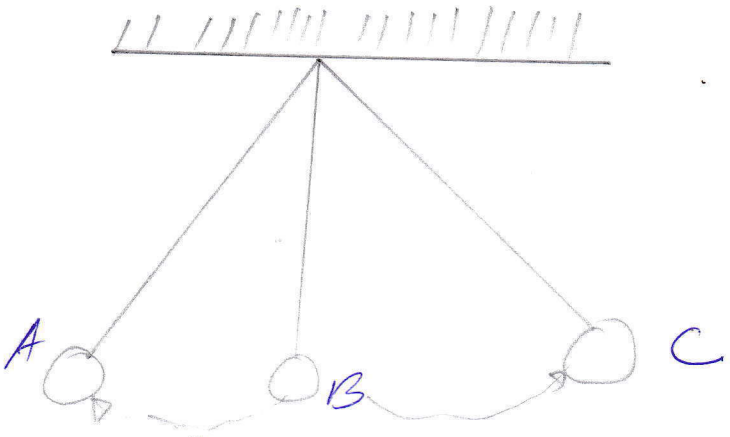
potential difference (U) = 10 V ——— 1 marks

Charge (φ) = ?

From $\varphi = CU$ ——— 1 marks

$= 250 \times 10 \times 10^{-6}$ F

06 (a) Consider the simple pendulum below.



1 mark

⇒ At point "A" the bob possess potential energy (P.E) which is maximum.
⇒ As a bob swings from "A" to "B" the energy changes from potential energy (P.E) to kinetic energy (K.E)
⇒ At point "B" the bob posses maximum kinetic energy (K.E-Emax) Since it is at the lowest position. (4 marks)
⇒ As a bob swings from point "B" to "C" its energy change from kinetic energy to potential energy (P.E)
⇒ At point C the bob again posses maximum potential energy (P.Emax) Since it is at the highest position.

06 (b) Data.

Mass (m) = 1000kg
Height (h) = 30m
Time taken (t) = 100sec (1 mark)
Gravitation force (g) = 10N/kg
Requires, power

Solution

$$\text{Power (P)} = \frac{\text{work done (w.d)}}{\text{Time (t)}}$$ (1 mark)

$$P = \frac{mgh}{t}$$

$$P = \frac{1000kg \times 10N/kg \times 30m}{100sec}$$ (1 mark)

06 (b) $P = \frac{1000 \times 10 \times 30}{100}$ — 1 marks

$= \frac{300000}{100}$

$= 3000\phi$

$P = 3000W$ — 1 marks

The power developed by the pump is 3000W.

---

7 (a) Archimedes principle state that

"When the body is totally or partially immersed in fluid it experiences an upthrust which is equal to weight of fluid displaced" by a body" (2 marks)

(b) Relative density

is a ratio of density of substance to the density of water ($R.D = \frac{\text{Density of substance}}{\text{Density of water}}$). (1 marks)

= Relative density has no SI unit. — (1 marks)

c. Data

Mass of the density bottle (M_1) = 15g

Mass of density bottle full filled with a fluid M_2 = 51g

Density of liquid / fluid = $1.2 g/cm^3$ — 1 marks

Volume of the bottle (V) = ?

From

$\text{Density of liquid} = \frac{\text{mass of liquid}}{\text{Volume of bottle}}$ — 1 marks

$\text{Volume of bottle (V)} = \frac{\text{mass of liquid}}{\text{Density of liquid}}$

$= \frac{M_2 - M_1}{\rho_L}$

$= \frac{51g - 15g}{}$ — 1 marks

7(c) $V = \frac{369 \times 10}{1.29 g/cm^3 \times 10}$

$V = 30 cm^3$

The volume of the bottle is $30 cm^3$ — (2 marks)

---

08. (a) Newton's second law of motion state that

" The rate of change of velocity of a body is proportional to the applied force and takes place in the direction of the force acts". (2 marks)

(b) Application of Newton's third law

(i) Rocket propulsion: the rate of ejection of exhaust gases is proportional to the force on the rocket.

(ii) On shooting using a gun, As the bullet leaves the gun, the body of the gun moves in the direction opposite to that of the bullet.

(iii) In kicking a ball, one gives the ball a forward force and in return the leg experiences some push due to the ball. (4 marks)

c. DATA GIVEN

Mass $m_1 = 8kg$

Initial velocity $(u_1) = 20 m/s$

Mass of another body $(m_2) = 4kg$

Velocity final before $(u_2) = 10 m/s$ ——— 1 mark

Velocity of body after collision $(v_1) = 15 m/s$

Required velocity of body another after (final velocity) $= v_2$

from

Principle of conservation of a linear momentum

$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 u_2$ ——— 1 mark

$8 \times 20 + 4 \times 10 =$

8 (c) $200 = 120 + 4V_2$ ——— 1 mark

$200 - 120 = 4V_2$

$\frac{80}{4} = \frac{4V_2}{4}$

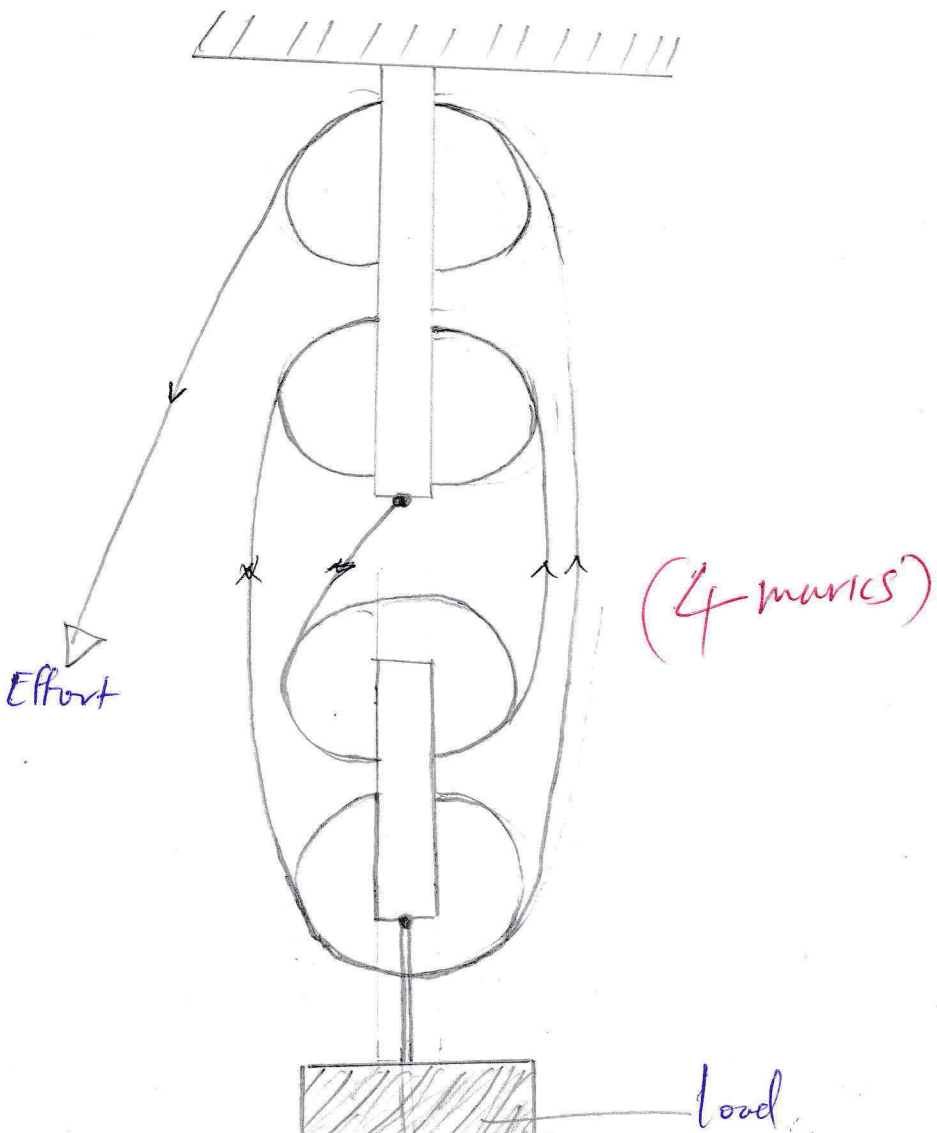
$V_2 = 20 m/s$

The final velocity is 20m/s ——— 2 marks

SECTION C (25 MARKS)

9. (a) An inclined plane makes it easier to raise a load because it reduce the force required to push the load up the plane, since it is just a component of weight. (3 marks)

(b)



(4 marks)

9 (c) DATA

V.R of a pulley system = number of pulleys = 8

Effort (E) = 200N

Efficiency of pulley system = 80% (1 marks)

mechanical Advatange = ?

maximu load = ?

from

$$\text{Efficiency of machine} = \frac{M.A}{V.R} \times 100\%$$

$$80\% = \frac{M.A}{8} \times 100\% \quad \text{—— 1 marks}$$

$$\frac{8 \times 80\%}{100\%} = \frac{100\% M.A}{100\%}$$

$$M.A = \frac{640\%}{100\%}$$

$$M.A = 6.4 \quad \text{—— 1 marks}$$

(ii) But Now

$$M.A = \frac{\text{Load}}{\text{Effort}} \quad \text{—— 1 marks}$$

$$6.4 = \frac{\text{load}}{200N}$$

$$\text{load} = 6.4 \times 200N$$

$$L = 1280N \quad \text{—— 2 marks}$$

The effort can raise a load of 1280N

---

10 (a) Ammeter in the circuit is always connected in series connection with other devices and voltmeter in the circuit is always connected in parallel connection with other devices

10(b) DATA

Voltage (V) = 2V

Resistance = 2Ω, 2Ω, 3Ω — 1 mark

Required Current I.

From

$$I = \frac{V}{R}$$

But R is connected in series — 1 marks

$R_T = R_1 + R_2$

$R_T = 2\Omega + 3\Omega$

$R_T = 5\Omega$

in parallel Connection

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$$ — 1 marks

$$\frac{1}{R_T} = \frac{1}{2\Omega} + \frac{1}{5\Omega}$$

$$\frac{1}{R_T} = \frac{5+2}{10}$$ — 1 marks

$$\frac{1}{R_T} = \frac{7}{10}$$

$R_T = 1.43\Omega$ — 1 marks

Since $I = \frac{V}{R}$ — 1 marks

$$I = \frac{2V}{1.43\Omega}$$

$I = 1.399A$ — 1 marks

The Ammeter reading is 1.399A